

PAPER_266: HUDF Primordial IGM Magnetic Field — UQFF Gravitational Meissner Effect and Superconducting Critical Boundary at $B_{\text{crit}} = 10^{11}$ T

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Abstract

The HUDFGalaxies MUGE equation contains the magnetic suppression factor $\text{corr}_B = 1 - B/B_{\text{crit}}$ with $B = 10^{-10}$ T (primordial intergalactic medium) and $B_{\text{crit}} = 10^{11}$ T as defined in the original C++ module header. This factor mirrors the behaviour of a Type II superconductor entering its upper critical field H_{c2} : at $B \rightarrow B_{\text{crit}}$, the UQFF gravitational field is expelled from the system, just as magnetic flux is expelled from a superconductor at H_{c2} .

The **uniquely rare discovery** of this paper is the identification of $B_{\text{crit}} = 10^{11}$ T as the **UQFF Gravitational Meissner Boundary** — above this threshold, the corr_B factor vanishes and UQFF gravity is completely quenched. The HUDF's ultra-weak primordial field $B = 10^{-10}$ T places it at $\text{corr}_B = 1 - 10^{-21} \approx 1$ (completely unquenched, fully-active UQFF). This represents the maximum possible UQFF gravitational activity — the HUDF is a cosmological benchmark for the **unquenched UQFF limit**. In contrast, neutron stars with surface fields $B \sim 10^{11}$ T (e.g., Cas A, PAPER_257) sit exactly at this critical boundary, explaining their anomalous UQFF behavior near the Force Equivalence Class transition.

1. System Parameters

Parameter	Symbol	Value	Units	Regime
HUDF primordial IGM	B_{HUDF}	10^{-10}	T	Fully unquenched
Neutron star surface	B_{NS}	$10^8\text{--}10^{12}$	T	Approaching/at boundary
Magnetar	B_{mag}	$10^{13}\text{--}10^{15}$	T	Above boundary (quenched)
UQFF critical	B_{crit}	10^{11}	T	Meissner boundary
QED Schwinger critical	$B_{\text{Schwinger}}$	4.4×10^{13}	T	QED pair-production

2. Gravitational Meissner Effect

2.1 The Magnetic Suppression Factor

The $\text{corr_B} = (1 - B/B_{\text{crit}})$ term controls the amplitude of UQFF gravitational acceleration:

$$U_{g4} = U_{g1} \cdot (1 - B/B_{\text{crit}}) = U_{g1} \cdot \text{corr}_B$$

The UQFF total:

$$U_{g,\text{UQFF}} = (U_{g1} + U_{g4}) \cdot (...) = U_{g1} \cdot \left(2 - \frac{B}{B_{\text{crit}}}\right) \cdot (...)$$

2.2 Phase Diagram by B/B_crit

System	B (T)	B/B_crit	corr_B	UQFF regime
HUDF primordial IGM	10^{-10}	10^{-21}	≈ 1.0	Fully active
Galaxy cluster (Faraday)	10^{-7}	10^{-18}	≈ 1.0	Fully active
Solar wind near Earth	5×10^{-9}	5×10^{-20}	≈ 1.0	Fully active
Neutron star (Cas A)	10^8	10^{-3}	0.999	Nearly active
PSRJ0030 pulsar	3×10^8	3×10^{-3}	0.997	Nearly active
B_crit boundary	10^{11}	1.0	0.0	QUENCH POINT
Strong magnetar	10^{13}	100	-99	Negative: unphysical

Note: $B > B_{\text{crit}}$ in C++ original has $\text{corr_B} < 0$ (unphysical). The validator in HUDFCriticalMagneticTerm permits $B \leq B_{\text{crit}} \times 1.1$ for probing the near-critical zone without full sign reversal.

2.3 Meissner Analogy

In Type II superconductors, the order parameter ψ (Cooper pair density) follows:

$$|\psi|^2 \propto \left(1 - \frac{B}{B_{c2}}\right) \quad \text{near } H_{c2}$$

The UQFF factor $\text{corr_B} = (1 - B/B_{\text{crit}})$ is **structurally identical** to this mean-field suppression, with corr_B playing the role of $|\psi|^2$. This suggests:

$$U_{g,\text{UQFF}} \propto |\psi|_{\text{UQFF}}^2 = 1 - B/B_{\text{crit}}$$

The UQFF gravitational quantum field is a condensate analogous to a superconducting order parameter. As $B \rightarrow B_{\text{crit}}$, the condensate melts — gravity quenches.

2.4 Critical Field Derivation

$B_{\text{crit}} = 10^{11}$ T corresponds to: - **Neutron star polar cap:** Standard pulsar surface field $B_s \sim 10^8\text{--}10^{12}$ T; at the critical boundary $B_s = B_{\text{crit}} = 10^{11}$ T, UQFF gravity is 99.9% active ($\text{corr}_B \approx 1 - B_s/B_{\text{crit}} \approx 0$ for $B_s = 10^{11}$ T) - **Landau level spacing:** $\hbar\omega_c = \hbar(eB/m_e c) = e\hbar B/m_e c$; at $B = 10^{11}$ T $\rightarrow \hbar\omega_c \approx 1.15 \times 10^{-2}$ J ≈ 72 MeV — near pion mass scale, suggesting B_{crit} marks the hadronic confinement-deconfinement boundary in QCD - **UQFF LENR coupling:** At $B = B_{\text{crit}}$, the 1.25 THz LENR oscillator resonance condition is modified by cyclotron resonance $\omega_{\text{LENR}} = \omega_c(B_{\text{crit}})$ for e^- at $B_{\text{crit}} \approx 1.25 \times 10^{12}$ rad/s $\rightarrow B \approx 7 \times 10^{-3}$ T. The mismatch confirms $B_{\text{crit}} = 10^{11}$ T is a separate, purely UQFF-gravitational boundary.

3. Gravitational Meissner Effect Theorem

Theorem (UQFF Gravitational Meissner Effect): The UQFF gravitational field $\mathcal{G}(B)$ obeys a Meissner-like suppression law:

$$\mathcal{G}(B) = \mathcal{G}_0 \cdot \left(1 - \frac{B}{B_{\text{crit}}}\right)$$

where $\mathcal{G}_0 = U_{g1}(2 - B/B_{\text{crit}}) \cdot (1 + f_{\text{TRZ}}) \cdot (1 + I(t))$ evaluated at $B = 0$.

The field $\mathcal{G}$ is **expelled** from the UQFF medium at $B = B_{\text{crit}}$ (gravitational quench), analogous to magnetic flux expulsion from a superconductor at H_c2 .

Corollary 1 (HUDEF Maximum): The HUDEF with $B_{\text{HUDEF}} = 10^{-10}$ T gives $\text{corr}_B = 1 - 10^{-21} \approx 1$, representing the **maximum UQFF gravitational activity achievable** in a cosmic environment. This makes the HUDEF the benchmark calibration field for fully-active UQFF.

Corollary 2 (NS Critical Zone): Neutron stars with $B \sim 10^{11}$ T sit at the Meissner boundary. PSRJ0030 and Cas A (PAPER_255, 257) with $B_0 \sim 10^8\text{--}10^9$ T are within 2-3 orders of B_{crit} — within the “upper critical zone” where UQFF is suppressed by 0.1-1%, explaining the slight deviation of their $F_{U_{Bi_i}}$ from the fulky unquenched theoretical maximum.

Corollary 3 (Magnetar Above-Critical): Magnetars with $B > B_{\text{crit}}$ would have $\text{corr}_B < 0$ — a physically distinct phase where U_{g4} contributes positively to gravity reversal, potentially explaining the anomalous braking indices of highly magnetised NSs.

4 Observational Predictions

- **HUDEF $z = 3.5$ ($B \approx 10^{-10}$ T):** $\text{corr}_B \approx 1.0 \rightarrow$ maximum $F_{U_{Bi_i}}$. ALMA Faraday rotation measurements of HUDEF background sources at $z > 3$ can constrain B_{HUDEF} and verify $\text{corr}_B \approx 1$.

- **Cas A neutron star ($B_0 \approx 10^5$ T thermal-scale):** From PAPER_257, $B_0 = 10^{-5}$ T $\rightarrow$ corr_B = $1 - 10^{-16} \approx 1$. Even with this tiny surface field in the PAPER_257 model, Cas A remains fully unquenched.
 - **Magnetar quench test:** An X-ray polarimetry observation of a magnetar with $B > 10^{11}$ T (e.g., SGR 1806-20, $B \sim 2 \times 10^{15}$ T) should show suppressed UQFF signature compared to the HUDF benchmark — a direct test of the Gravitational Meissner Effect.
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5. References

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